ICT Level 2

Teaching Guide

Level 2 - Class 7

Level: Level 2

Class: Class 7

Duration: June - January (8 months)

Period: 40 minutes

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Table of Contents

Curriculum Overview

ICT Level 2 builds on the foundations established in Level 1, designed for Class 7 students in residential schools. The curriculum follows the 5E instructional model and integrates Singapore's Concrete-Pictorial-Abstract approach for deeper conceptual understanding.

Themes Covered

Chapter	Theme	Periods	Key Topics
1	Networking and Internet	12	Network types, LAN/WAN, Internet services, Email
2	Advanced Graphics	12	Image editing, Layers, Filters, GIMP
3	Multimedia	12	Audio/Video editing, Presentation tools
4	Data Handling	12	Spreadsheets, Charts, Formulas
5	Programming with Scratch	12	Variables, Conditions, Loops
6	Digital Citizenship	12	Online safety, Cyberbullying, Digital footprint

Pedagogical Approach

5E Model Implementation

Phase	Duration	Activities
Engage	5 min	Real-world connection, regional metaphors, questions
Explore	10 min	Hands-on activity, Driver-Navigator pairs, Lab work
Explain	10 min	Live coding, Concept introduction, C-P-A approach
Elaborate	10 min	Extended activity, Parson's problems, Application
Evaluate	5 min	Formative check, Exit tickets, Self-assessment

International Pedagogies Used

- **Singapore C-P-A:** Concrete manipulatives to Visual blocks to Abstract code
- **UK NCCE:** Unplugged Computing, Live Coding, Parson's Problems
- **Driver-Navigator:** Pair programming for lab sessions
- **Rural Scaffolding:** Regional metaphors (grain sorting, bus routes, hostel inventories)

Lesson Plan Structure (40 Minutes)

Time	Phase	Teacher Activity	Student Activity
0-5 min	Engage - Retrieval & Hook	Present real-world scenario using regional metaphor	Discuss, share observations
5-15 min	Explore - Guided Discovery	Guide hands-on exploration in pairs	Work in Driver-Navigator pairs, explore software
15-25 min	Explain - Modelling	Live coding demonstration, introduce concepts	Follow along, ask questions, take notes
25-35 min	Elaborate - Application	Assign extension activity or Parson's problem	Apply learning, solve problems independently
35-40 min	Evaluate - Hinge Q & Exit	Conduct formative assessment, assign homework	Complete exit ticket, receive dormitory task

Anatomy of a Lesson Plan (v2, August 2026)

Every lesson plan PDF now follows this fixed anatomy. Teachers should read top-to-bottom once before class; the plan is print-ready in black-and-white.

- **Header bar** - Lesson number, topic (italic), duration (40 min), and an archetype badge: CONCEPT (5E inquiry), LAB (Rosenshine I-do/We-do/You-do), or DEBUG (PRIMM Predict-Run-Investigate-Modify-Make). Chapters 2-3 are tagged LAB; chapters 5-6 DEBUG.
- **Success Criteria** - 'I can…' statements derived from objectives; used again during Evaluate for self-assessment ticks.
- **Resources Needed** - Placed at the TOP so materials can be gathered before class. Split mentally into Lab Kit vs Library Paper Pack for the 25/25 station rotation.
- **Key Vocabulary** - Four terms with one-line definitions for EAL learners; pre-teach during Engage.
- **Five phases** (5/10/10/10/5) - Each phase shows TEACHER and STUDENTS blocks side-by-side by left rule colour (blue/green).
- **Differentiation** - Four tiers: Approaching / On Level / Advanced / SEND.
- **Formative Assessment** - Method plus the Computational Thinking skill assessed (maps to the CT rubric in assessments).
- **Hinge Question** - Embedded in Evaluate; if fewer than 80% answer correctly, reteach 3 minutes before Elaborate.
- **Dormitory Task** - Interleaved retrieval: 3 facts from today, 2 connections to previous chapter, 1 question, plus one focused task and a weekly spiral recall.
- **Colophon** - Author, license (CC BY-SA 4.0), version.

Assessment Framework

Assessment Components (Per Class)

Assessment Type	**Count**	**Marks Each**	**Total**
Formative (FA-1 to FA-4)	4	30	120
Unit Tests	4	10	40
Summative (SA-1, SA-2)	2	40	80
Practical Lab Exams	2	20	40
Grand Total	**12**		**280**

Resources Required

Hardware

- Computer lab with 1 PC per 2 students (minimum)
- Internet connection (2 Mbps minimum)
- Projector for demonstrations
- Speakers and headphones

Software (Free/Open Source)

- Operating System: Ubuntu Linux or Windows
- Office Suite: LibreOffice / Google Workspace
- Graphics: GIMP, Inkscape
- Audio: Audacity
- Video: Kdenlive
- Programming: Scratch
- Browser: Firefox / Chromium

Dormitory Tasks (Zero Lab Access)

Lesson plans now embed interleaved 3-2-1 retrieval as the standard dormitory task format. For residential schools with limited lab time, these paper-and-pencil tasks can be assigned for supervised evening study:

- **Code Dry-Running:** Trace Scratch programs on paper
- **Variable State Tables:** Track variable values through program execution
- **Flowchart Drawing:** Create flowcharts for algorithms
- **Network Diagrams:** Draw and label network topologies on paper
- **Spreadsheet Planning:** Design spreadsheet layouts with formulas on paper
- **Binary Conversion:** Practice decimal to binary conversions
- **Cyberbullying Scenarios:** Discuss and write responses to online safety scenarios
- **Digital Footprint Journal:** Reflect on personal online activities

Capstone Projects

Project 1: School Network Design

Integration: Science + ICT

Description: Students design a network layout for a hypothetical school campus, identifying devices, connections, and resource sharing requirements. They create network diagrams and present their proposals.

Skills: Network topologies, LAN/WAN concepts, diagramming, presentation

Project 2: Community Data Dashboard

Integration: Mathematics + ICT

Description: Students collect data about their community (population, rainfall, agricultural produce) and create a spreadsheet with charts and formulas to visualize and analyze the information.

Skills: Spreadsheet formulas, chart creation, data analysis, digital presentation

Unplugged Activities Library

Activity 1: Network Relay (Ch 1 - Networking)

Duration: 20 minutes | **Materials:** Paper, tape

Steps:

1. Students form a “network” - some are computers, some are routers, some are servers
2. Pass messages (paper slips) from one “computer” to another through “routers”
3. If a “router” is busy (student sits down), messages must find another path
4. Discuss: What happens when parts of a network fail?

CT Skill: Network thinking, redundancy, decomposition

Activity 2: Layer Editing Puzzle (Ch 2 - Graphics)

Duration: 15 minutes | **Materials:** Tracing paper, colored pencils

Steps:

1. Draw a simple landscape on one sheet (background)
2. On tracing paper, draw only the tree (layer 1)
3. On another tracing paper, draw only the house (layer 2)
4. Stack them. Discuss: Why do graphic artists use layers?

CT Skill: Abstraction, decomposition

Activity 3: Audio Editing Cards (Ch 3 - Multimedia)

Duration: 15 minutes | **Materials:** Printed story cards

Steps:

1. Give groups a story told in 10 cards

2. Task: Remove 2 cards that are not essential to the story
3. Task: Reorder 3 cards to make the story more interesting
4. Discuss: How is this like editing audio/video?

CT Skill: Abstraction, pattern recognition

Activity 4: Algorithm Recipe (Ch 5 - Programming)

Duration: 15 minutes | **Materials:** Paper, pencil

Steps:

1. Write a recipe for making chai - but use only these words: START, STOP, REPEAT, IF, THEN
2. The recipe must include: boiling water, adding tea leaves, adding milk, checking taste
3. Trade with another group and follow their recipe exactly
4. Discuss: What happens if steps are missing or unclear?

CT Skill: Algorithm design, debugging

Activity 5: Digital Footprint Audit (Ch 6 - Digital Citizenship)

Duration: 20 minutes | **Materials:** Paper, pencil

Steps:

1. List every app, website, and social media you use
2. For each, write: What information does it have about you?
3. Draw a “footprint” showing all your digital traces
4. Discuss: Who can see all this information?

CT Skill: Abstraction, evaluation

Cross-Curricular Integration Map

Mathematics Connections

- **Ch 1:** Network protocols use binary numbers and addressing
- **Ch 2:** Color theory involves ratios and percentages
- **Ch 3:** Audio/video editing uses timelines (number line concept)
- **Ch 4:** Data handling uses statistics (mean, median, mode, range)
- **Ch 5:** Programming variables are algebraic expressions
- **Ch 6:** Digital ethics involves logical reasoning (IF-THEN)

Science Connections

- **Ch 1:** Electromagnetic waves carry network signals (Physics)
- **Ch 2:** Light and color perception (Biology/Physics)
- **Ch 3:** Sound waves and human hearing (Physics/Biology)
- **Ch 4:** Data collection follows scientific method
- **Ch 5:** Algorithms model chemical reactions and biological processes
- **Ch 6:** E-waste and environmental science

Language Connections

- **Ch 1:** Technical writing: network documentation
- **Ch 2:** Visual literacy: interpreting infographics
- **Ch 3:** Script writing for presentations
- **Ch 4:** Data description and report writing
- **Ch 5:** Code comments and documentation
- **Ch 6:** Persuasive writing on digital ethics topics

Digital Citizenship Module

Class 7: Being Responsible Online

Duration: 3-4 lessons (integrated throughout the year)

Lesson: Password Security

- Create passphrases: “MyHostelHas20Rooms!” is better than “password123”
- Practice creating strong passwords for different accounts
- Rule: Different accounts need different passwords

Lesson: Identifying Misinformation

- Analyze 3 WhatsApp forwards: which are real, which are fake?
- Check: Source, date, multiple sources confirming, reverse image search
- Rule: “Verify before you share”

Lesson: Cyberbullying Response Plan

- Create a 3-step response: Ignore -> Block -> Report to adult
- Practice role-playing scenarios
- Create a class “Safety Pledge” poster

Lesson: Digital Footprint Management

- Google yourself or your school. What information is public?
- Think before you post: Would you show this to your grandparents?
- Rule: “Your digital footprint is your permanent record”